

**CYBER SECURITY BEHAVIOUR AMONG THE COLLEGE STUDENTS IN
COIMBATORE DISTRICT**

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Abstract:

(The present study has been carried out to analyze the behaviour of college students towards cyber security. The study themes on analyzing the behaviour of college students with regard to cyber knowledge, security implications and cyber threats. The research study uses both primary and secondary sources of data. Stratified random sampling technique was used to collect the data from 300 college students of medical, engineering, arts and science streams. Factor analysis and confirmatory factor analysis was used for analyzing the data collected on the behavioural aspect. The data reduction technique has extracted three factors namely cyber knowledge, security implications and cyber threats. These factors were confirmed using confirmatory factor analysis.)

Keywords: Cyber, Threats, Knowledge. Security

Introduction

Cyber security is concerned with making cyberspace safe from various threats, called cyber-threats. The notion of “cyber-threats” is rather vague and implies the venomous use of information and communication technologies (ICT) either as a target or as a tool by a wide range of evil-minded actors. Cyber security refers to multi fold activities and other measures, both technical and non-technical, intended to protect computers, computer networks, related hardware and devices software, and the information they contain and communicate, including software and data, as well as other elements of cyberspace, from all threats, including threats to the national security. It also measures the degree of protection resulting from the application of these activities and measures.

Cyber security is thus more than just information security or data security and also relies on digital access, digital literacy, digital communication, commerce, digital etiquette and digital health and wellness. Thus, the information security lies at the heart of the matter. India has been the second most cyber attacks affected during 2016 to 2018, according to a new Data Security Council of India (DSCI). On turn India’s ban on 59 Chinese apps was only the tremendous start and there will be greater scrutiny of companies, device makers and apps with exposure to China, which can trigger retaliatory attacks. Since we have permitted the Chinese for so many years to invest in critical infrastructure, especially in communication. According to the Internet and Mobile Association of India (IMAI), there are currently 580 million internet users with a growth rate of roughly 20% in India.

India estimates the internet population would be 70 crore during 2019. India is among the top 10 countries facing cyber-attacks. These incidents have increased manifold during the lockdown – almost three times increase in case of spamming, and scanning of ICT systems here is significant increase in incidents relating to hacking. The country has estimated 526 million active internet users and of which 18 per cent of the users fall in the age group of 16 to 19 years of age.

College students are not as worried as they should be about the threat of identity theft (Farzan, A, 2015). India’s vast size limits the physical solution to many problems and digital solutions are thus helping in solving many human and economic problems. Hence, the students in today’s era are much blinded and bounded towards the usage of various online activities and applications. Thus, the present study aims to empirically analyze the behaviour towards cyber security among the college students in Coimbatore District.

Review of Literature

Farzana Quayyum, Daniela S. Cruzes and Letizia Jaccheri (2021) have stated in their study that Cyber security is a methodology to educate internet users to be sensitive to the various cyber threats and the vulnerability of computers and data to these threats and discussed about the importance of information security and their responsibilities to exercise sufficient levels of information control to protect the organization's data and networks. In this study the interview method and experiment action research method was taken and reveals the importance of cyber security awareness Based on the study cyber security awareness has two primary purposes: alerting the internet users about cyber security risks and enhancing the internet users' understanding of cyber security risks to be sufficiently committed to embracing security during internet use. The internet users raise user awareness about cyber security and privacy issues.

Ulrik Franke and Joel Brynielsson (2014) Cyber situational awareness is attracting much attention. It features prominently in the national cyber strategies of many countries, and there is a considerable body of research dealing with it. However, until now, there has been no systematic and up-to-date review of the scientific literature on cyber situational awareness. This article presents a review of cyber situational awareness, based on systematic queries in four leading scientific databases. 102 articles were read, clustered, and are succinctly described in the paper. The findings are discussed from the perspective of both national cyber strategies and science, and some directions for future research are also examined.

Samaher Al-Janabi and Ibrahim Al-Shourbaji (2016) in their study stated that Information security awareness can play an important role in facing cyber-attacks by intruders. The main goal of this paper is to analyze the information security awareness among academic staff, researchers, undergraduate students and employee within educational environments in the Middle East in an attempt to understand the level of awareness of information security, the associated risks and overall impact on the institutions. The results reveal that the participants do not have the requisite knowledge and understanding of the importance of information security principles and their practical application in their day-to-day work. This situation can however be corrected through comprehensive awareness and training programs as well as adopting all the necessary safety measures at all levels of the institution to ensure that the students, academic staff and employees are trustworthy, technology savvy and keep their data safe. Without such training programs and awareness, there will be negative consequences on IT systems and their application usage, as well as on users' personal security now and in the future. From the weaknesses identified in this survey, some essential recommendations are put forward to remedy the situation.

Nabin Chowdhury and Vasileios Gkioulos (2021) this study seek to establish the current state-of-the-art in cyber-security training offerings for critical infrastructure protection and the key performance indicators (KPIs) that allow evaluating their effectiveness. Training was given to internet users based on simulation methods were identified as the optimal training method. After the training, evaluation requires the use of performance indicator. The identified sources have been examined according to a formal data extraction form, allowing the analysis of relevant training solutions, methodologies, target groups and focus areas. The results show that solutions that provide hands-on experience, team skills development, high level of real-life fidelity are often preferred to other options, with simulation-based solutions showing the highest amount of research and development.

Saroj Mehta & Vikram Singh (2013) the increased reliance of individuals/organizations on cyberspace has resulted in to a corresponding increase in the cybercrimes. Coupled with lack of proper training and education, the low level of awareness of the Indian society about the cybercrime has resulted into a spurt of cybercrimes. At times, even the law enforcement officers do not have proper training and other requisite expertise for tackling cybercrime. India may succeed in combating the problem of cybercrimes by adopting a synergetic approach wherein technological measures and proper legislative framework with a properly trained human resource in a tech-savvy society. The study aimed to collect responses with regards to the knowledge and awareness of respondents

towards cyber laws in India. A three point structured questionnaire was designed to find the results. Such data were collected from the students/teachers and employee. An individual participant constituted the sampling unit whereas probability sampling. The bases of findings of present work it could be coined that there is a significant difference between the awareness level of male and female users of internet services

Dr. Seema Girdhar, Rajat Saxena (2021) In the paper researchers has tried to study on awareness level of youth on cyber security. For this both close and open ended questionnaires are framed in order to collect the information and analyze the awareness level of youth on cyber security. Youth of capital city Delhi are targeted for gathering the information on cyber security. The Exploratory Research Design has been adopted by focusing on burning issue of cyber security in Internet world. Non Probabilistic sampling method has been used and in that convenience and judgmental sampling approach was adopted. Little awareness on the part of youth could save the data and also helps in leaving tension free life. Although the security awareness field is still very immature, the changing of people behavior towards security can be achieved by using soft skills such as communication and marketing, initiatives like email and posters can be used to endure a positive security culture.

Problem Statement

In the modern and digitalized world the usage of digital devices and applications has become a part of life. The higher education students are now much bounded on using various applications on learning, self grooming, commerce, shopping, wellness and even more. Day by day the usages on these apps are increasing and the security issues have never been focused by the students. The out focusing on these apps may lead to cyber threat and tends to misbehaviour also. The cyber threat will lead on lack of knowledge and behaviour. Knowledge on cyber security will enhance the behaviour of college students. Hence, cyber threat is the major issue for digital India. The issue on cyber threat can be controlled by raising the awareness, knowledge and behaviour of the college students. The present study aims to answer

1. How would the college students behave on the cyber security hazards?

Objective

The objective of the study is

1. To analyze on the behaviour on cyber security among the youths (college students) in Coimbatore District.

Methodology

Research methodology describes the magnitude of the research study identified, the reason behind framing the hypotheses, and to collect the data. The design of the study elaborates the descriptive type of research which utilizes the survey method and mentions about the method of data collection. The Data for the study is being carried out from both primary and secondary sources for analyzing the behaviour of the college students. The primary data is collected from the students perusing their higher education at the graduate level using a well structured questionnaire. The study was conducted during the pandemic Month September, 2020. The data collection during the period will enhance the knowledge and skills of the students, as they are in virtual learning situation, which in turn their use of devices will be a part and parcel in their day to day activities. The secondary sources of data will be collected from journals, news papers, textbooks, internet sources, etc.

The study utilizes 300 sample respondents using stratified proportionate random sampling technique. The streams of the students are fixed as strata. These data's were analyzed using Exploratory Factor Analysis and Confirmatory Factor Analysis.

Discussions

Exploratory Factor Analysis

The exploratory factor analysis explores about the behavior of college students on cyber security.

Table – 1 KMO and Bartlett's Test

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.820
Bartlett's Test of Sphericity	Approx. Chi-Square	9887.815
	Df	171
	Sig.	.000

Source: Compiled and Calculated from Primary Data (SPSS Output)

The above table reveals that the (KMO) Kaiser-Meyer-Olkin measure of sampling adequacy is 0.820 and Bartlett's test of sphericity significant value is 0.000 at 5 per cent level of significance. It becomes clear that the factors are significant and data reduction can be employed to explore the components.

Table-2 Communalities

	Initial	Extraction
I give a great priority on using my mobile devices securely	1.000	.645
I am much concerned that cyber security could affect personal safety	1.000	.547
It is much important to renew or update your device passwords	1.000	.718
I am very much concerned about strengthening the passwords	1.000	.711
Using of internet would compromise my personal information	1.000	.707
I am aware on security implications of https: versus http: in website address	1.000	.667
I am much concerned on phishing attacks and their dangers	1.000	.698
Usage of mobile devices regularly could result in bad opinion	1.000	.421
Clicking on unknown links provided in email would result in good practice.	1.000	.623
Using a credit card for shopping online is secure	1.000	.710
I am much concerned about identity and privacy when using social media	1.000	.571
I am very careful when posting the location or photos on social media	1.000	.672
I am much concerned that smart phone can get viruses	1.000	.670
I had a wide knowledge on installing the apps on device securely	1.000	.444
I am very careful on opening e mails from unfamiliar senders	1.000	.762
I am familiar on potential viruses/malwares on your devices	1.000	.792
I am concerned on connecting to an open wi-fi	1.000	.873
I am careful while downloading files from the Internet	1.000	.650
I am updating the devices regularly with cyber knowledge	1.000	.678

Extraction Method: Principal Component Analysis.

Source: Compiled and Calculated from Primary Data (SPSS Output)

Communalities are referred as the sum of square loading of the variables. The communality value should be higher than the threshold value (0.3). The above table states that the communality value ranges between 0.421 and 0.873 and all the 19 components are above the threshold limit of 0.3. Hence all the statements are said to be significant statistically.

Table – 3 Total Variance Explained

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings ^a
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %	Total
1	8.226	43.294	43.294	8.226	43.294	43.294	6.826
2	2.970	15.634	58.928	2.970	15.634	58.928	6.440
3	1.363	7.175	66.103	1.363	7.175	66.103	4.775
4	1.267	6.671	72.774				
5	.882	4.643	77.417				
6	.703	3.702	81.119				

7	.557	2.931	84.050			
8	.543	2.856	86.906			
9	.463	2.435	89.341			
10	.378	1.990	91.331			
11	.369	1.942	93.273			
12	.263	1.382	94.655			
13	.220	1.157	95.812			
14	.207	1.087	96.899			
15	.161	.847	97.746			
16	.149	.786	98.532			
17	.119	.625	99.157			
18	.092	.483	99.639			
19	.069	.361	100.000			

Extraction Method: Principal Component Analysis.

Source: Compiled and Calculated from Primary Data (SPSS Output)

The above table explains about the cyber security behaviour which is extracted using principal component analysis and rotation with normalization. It is done with Promax and 3 components are extracted from 19 items. The cumulative per cent of sum of squared loadings is extracted to 66.10%. The eigen values are above the threshold limit of 1 is found in all the three factors and the values are 8.226, 2.970 and 1.363.

Table – 4 Factor Analysis

Factors	Statements	Component		
		1	2	3
Cyber Knowledge	I am much concerned on phishing attacks and their dangers	.881		
	I am careful while downloading files from the Internet	.846		
	I give a great priority on using my mobile devices securely	.810		
	I am aware on security implications of https: versus http: in website address	.760		
	I am much concerned that cyber security could affect personal safety	.663		
	I am updating the devices regularly with cyber knowledge	.662		
	Using of internet would compromise my personal information	.635		
Security Implications	Using a credit card for shopping online is secure		.907	
	It is much important to renew or update your device passwords		.725	
	I am much concerned about identity and privacy when using social media		.706	
	I am very much concerned about strengthening the passwords		.706	
	Clicking on unknown links provided in email would result in good practice.		.687	
	Usage of mobile devices regularly could result in bad opinion		.593	
	I had a wide knowledge on installing the apps on device securely		.544	
Cyber Threats	I am very careful on opening e mails from unfamiliar senders			.907
	I am concerned on connecting to an open wi-fi			.883
	I am very careful when posting the location or photos on social media			.866

	I am familiar on potential viruses/malwares on your devices			.778
	I am much concerned that smart phone can get viruses			.761
Extraction Method: Principal Component Analysis.				
Rotation Method: Promax with Kaiser Normalization.				
a. Rotation converged in 5 iterations.				

Source: Compiled and Calculated from Primary Data (SPSS Output)

The factor analysis table infers about the cyber security behaviour of college students. The first factor entitled “**Cyber Knowledge**” explores the following seven items such as, I am much concerned on phishing attacks and their dangers, I am careful while downloading files from the Internet, I give a great priority on using my mobile devices securely, I am aware on security implications of https: versus http: in website address, I am much concerned that cyber security could affect personal safety, I am updating the devices regularly with cyber knowledge and Using of internet would compromise my personal information.

The “**Security Implications**” is the second factor inspect the following items such as, Using a credit card for shopping online is secure, It is much important to renew or update your device passwords, I am much concerned about identity and privacy when using social media, I am very much concerned about strengthening the passwords, Clicking on unknown links provided in email would result in good practice and Usage of mobile devices regularly could result in bad opinion.

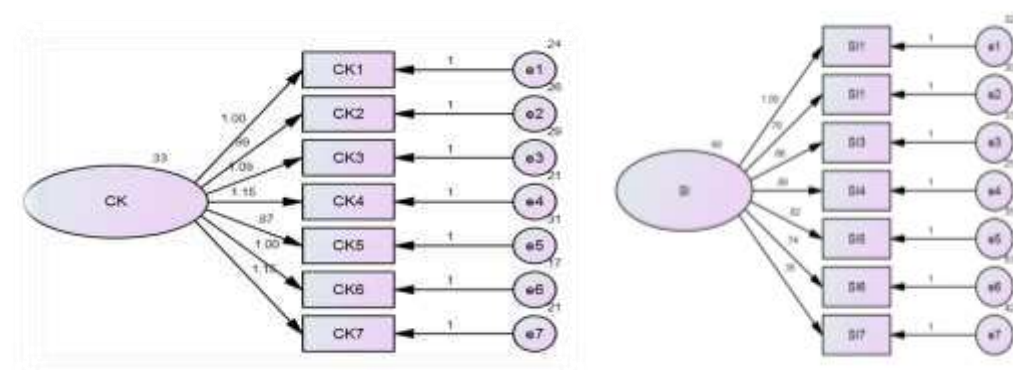
The third factor “**Cyber Threats**” explores the following items, I am very careful on opening e mails from unfamiliar senders, I am concerned on connecting to an open wi-fi, I am very careful when posting the location or photos on social media, I am familiar on potential viruses/malwares on your devices and I am much concerned that smart phone can get viruses.

Confirmatory Factor Analysis

Confirmatory factor analysis has been used to confirm the factors which were extracted by data reduction technique. Confirmatory factor analysis has been done for cyber knowledge, security implications and cyber threats.

The fit indices GFI value stands as 0.885, 0.830 and 0.948 also satisfies the threshold limit of greater than 0.90 and 0.80 to 0.90. The AGFI ranges from 0.880, 0.956 and 0.844, have also satisfied the suggested value of 0.90 and 0.80 to 0.90. The RMSEA values of the constructs are calculated as 0.052, 0.058, and 0.051 satisfies the suggested limit of less than 0.05 and 0.08 to 0.05. The NFI indices are 0.950, 0.953 and 0.953 The CFI indices are stood at 0.953, 0.925, and 0.955 satisfies the threshold limit of above 0.90 and 0.80 to 0.90. Thus, these constructs and statements can be considered for the structural model.

Figure. I Confirmatory Factor Analysis for Cyber Knowledge, Security Implications and Cyber Threats.



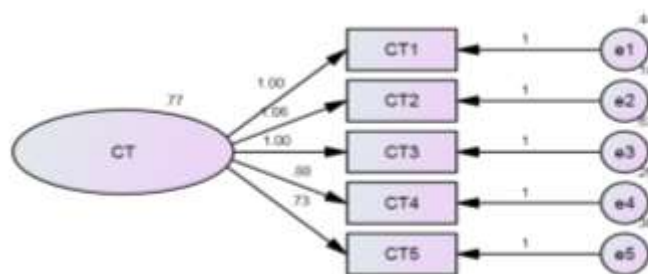


Table -5 Fit Indices Obtained

Measures	Suggested Limits	Obtained Value		
		Cyber Knowledge	Security Implications	Cyber Threats
CMIN/DF	<3.0 ^a , 3 to 5 ^b	2.130	3.854	2.909
P – Value	>0.5	.000	.000	.000
GFI	>0.90 ^a , >0.80 ^b	.885	.830	.948
AGFI	>0.90 ^a , >0.80 ^b	.880	.956	.844
RMSEA	<0.05 ^a , <0.08 ^b	.052	.058	.051
NFI	>0.90 ^a , >0.80 ^b	.950	.953	.953
CFI	>0.90 ^a , >0.80 ^b	.953	.925	.955
Note: a- Good Fit, b- Acceptable Fit		Cyber Knowledge (CK), Security Implications (SI), Cyber Threats (CT)		

Source: Compiled and Calculated from Primary Data (AMOS Output)

Conclusion

The knowledge on cyber security keens on protecting the sensitive data, softwares and network from cyber attacks. While there are so many clever practices to avoid cyber attacks and it is a must to protect our own data from threats. Hence the change and adoption of behaviour towards gaining cyber knowledge, concern on security implications and cyber threat strong and safe practices would mitigate the risk of a cyber attack. Now a day college students are playing a heroic role in fighting against cyber attacks. The college students are more aware on cyber threats and are involved in secured practices while using their devices.

Reference

1. Dr. Seema Girdhar, Rajat Saxena & Ankshit Singhal. (2021), ISSN 0975-1874- Youth Awareness towards Cyber Security: An Exploratory Study- *CPJ-GLOBAL* – Vol.XIII, pp.64-70
2. Farzana Quayyum Daniela S. Cruzes Letizia Jaccheri (2021) “Cyber security awareness for children: A systematic literature review”, *International Journal of child computer interaction*, Vol.30, Pp. 1-25.
3. Goodman Symour E and Herbert S. Towards a Safer and More Secure Cyberspace. National academies Press, 2007.
4. Nabin Chowdhury Vasileios Gkioulos (2021)Cyber security training for critical infrastructure protection: *A literature review*, Vol.40,- ELSEVIER
5. Paulson LD. Spam hits instant messaging. *Computer and Internet Security*, 37 no 4:18, 2004.
6. Samaher Al-Janabi and Ibrahim Al-Shourbaji (2016). A Study of Cyber Security Awareness in Educational Environment in the Middle East, *Journal of Information & Knowledge Management*, Vol. 15(1), 1650007
7. Saroj Mehta & Vikram Singh Department of Computer Science Chaudhary Devi Lal University, Sirsa (2013). A study of awareness about cyber laws in the Indian society. *International Journal of Computing and Business Research (IJCBR)*, Vol(4) 2229-6166-
8. Tygar J Rachna D and Marti Hearst. Proceedings of the conference on human factors in computing systems. In *Why Phishing Works*, 2006.
9. Ulrik Franke* , Joel Bry nielsson (2014)- Cyber situational awareness A systematic review of the literature, *journal homepage: www.elsevier.com/locate/cose-Pg no1-14*